

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I ___N.Umasri(192424240)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Umasri

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2	3	2	1	1	9
Q2	2	3	2	1	1	9
Q3	2	3	2	1	1	9
Q4	2	2	2	1	1	8

Faculty Signature

Experiments

QUESTIONS:4 Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)

ACK

The image shows a Wireshark capture of a TCP connection setup. The main packet list on the left shows three packets:

- Packet 2: SYN from 10.187.165.146 to 23.33.114.104, port 443.
- Packet 3: SYN-ACK from 23.33.114.104 to 10.187.165.146, port 443.
- Packet 4: ACK from 10.187.165.146 to 23.33.114.104, port 443.

The packet details pane on the right shows the details of the selected packet (Packet 4, ACK). It indicates that the acknowledgment number is 1, which corresponds to the SYN packet's sequence number.

Syn

The image shows a Wireshark capture of a TCP connection setup. The main packet list on the left shows three packets:

- Packet 1: SYN from 13.107.6.158 to 10.187.165.146, port 443.
- Packet 2: SYN-ACK from 10.187.165.146 to 13.107.6.158, port 443.
- Packet 3: ACK from 10.187.165.146 to 13.107.6.158, port 443.

The packet details pane on the right shows the details of the selected packet (Packet 3, ACK). It indicates that the acknowledgment number is 0, which corresponds to the SYN packet's sequence number.

Ack&syn

The image shows a Wireshark capture of a network packet. The main pane displays a list of packets, with packet 33 selected. The packet details pane on the right shows the structure of the selected packet, which is a Transmission Control Protocol (TCP) segment. The packet is an acknowledgment (ACK) with a sequence number of 1 and a length of 0. The source port is 443 and the destination port is 57790. The packet is captured on interface \Device\NPF_{6F5A5BE-5594-41...}.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	13.107.6.158	10.187.165.146	TLsv1.2	693	Application Data
2	0.043782	10.187.165.146	13.107.6.158	TCP	54	61343 → 443 [ACK] Seq=1 Ack=640 W
28	2.988802	10.187.165.146	35.190.26.192	TCP	54	55405 → 443 [FIN, ACK] Seq=1 Ack=
29	2.989733	10.187.165.146	35.187.148.146	TCP	66	57790 → 443 [SYN] Seq=0 Win=65535
31	3.174686	35.190.26.192	10.187.165.146	TCP	54	443 → 55405 [FIN, ACK] Seq=1 Ack=
32	3.174621	10.187.165.146	35.190.26.192	TCP	54	55405 → 443 [ACK] Seq=2 Ack=2 Win
33	3.175253	35.187.148.146	10.187.165.146	TCP	66	443 → 57790 [SYN, ACK] Seq=0 Ack=
34	3.175402	10.187.165.146	35.187.148.146	TCP	54	57790 → 443 [ACK] Seq=1 Ack=1 Win
35	3.177642	10.187.165.146	35.187.148.146	TLsv1.2	2082	Client Hello
39	3.379236	35.187.148.146	10.187.165.146	TCP	66	[TCP Window Update] 443 → 57790
40	3.379236	35.187.148.146	10.187.165.146	TCP	56	443 → 57790 [ACK] Seq=1 Ack=2029
41	3.379236	35.187.148.146	10.187.165.146	TLsv1.2	1334	Server Hello

Frame 33: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{6F5A5BE-5594-41...}

Ethernet II, Src: 12:18:fd:44:5a:94 (12:18:fd:44:5a:94), Dst: 28:2e:89:cd:a8:3e (28:2e:89:cd:a8:3e)

Internet Protocol Version 4, Src: 35.187.148.146, Dst: 10.187.165.146

Transmission Control Protocol, Src Port: 443, Dst Port: 57790, Seq: 0, Ack: 1, Len: 0

Source Port: 443
Destination Port: 57790
[Stream index: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 799135021
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)

0000 28 2e 89 cd a8 3e 12 18 fd 44 5a 94 00 00 45 00 (.....DZ...E
0010 00 34 00 00 00 00 33 06 df 29 23 bb 94 92 0a bb 4..@3..)#....
0020 a5 92 01 bb e1 be 2f a1 d5 2d b5 9d 87 37 80 12/..:~7..
0030 ff 28 e2 bf 00 00 02 04 05 14 01 01 04 02 01 03 -(.....).....
0040 03 07 ..

2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size

with TCP and note the absence of sequence numbers or acknowledgments.

Discuss how this lightweight design impacts speed and reliability. (BL4)

(10)

Dns

The image shows a Wireshark capture of a network packet. The main pane displays a list of packets, with packet 530 selected. The packet details pane on the right shows the structure of the selected packet, which is a User Datagram Protocol (UDP) segment. The packet is a Domain Name System (DNS) query with a source port of 49664 and a destination port of 53. The packet is captured on interface \Device\NPF_{6F5A5BE-5594-4BC0-86FF-816436...}.

No.	Time	Source	Destination	Protocol	Length	Info
154	24.075103	10.187.165.146	10.187.165.253	DNS	80	Standard query 0x73e4 A beacons.gcp.gvt2.com
157	24.112080	10.187.165.253	10.187.165.146	DNS	167	Standard query response 0x5a0d HTTPS beacons.gcp.
158	24.112080	10.187.165.253	10.187.165.146	DNS	126	Standard query response 0x73e4 A beacons.gcp.gvt2
348	59.137434	10.187.165.146	10.187.165.253	DNS	71	Standard query 0xaafb HTTPS chatgpt.com
349	59.137992	10.187.165.146	10.187.165.253	DNS	71	Standard query 0xd163 A chatgpt.com
354	59.208614	10.187.165.253	10.187.165.146	DNS	103	Standard query response 0xd163 A chatgpt.com A 10
363	59.936138	10.187.165.253	10.187.165.146	DNS	132	Standard query response 0xaafb HTTPS chatgpt.com
529	72.801587	10.187.165.146	10.187.165.253	DNS	79	Standard query 0xcdc8 HTTPS clients4.google.com
530	72.802056	10.187.165.146	10.187.165.253	DNS	79	Standard query 0xc157 A clients4.google.com
535	73.016907	10.187.165.253	10.187.165.146	DNS	153	Standard query response 0xcdc8 HTTPS clients4.goo
536	73.016907	10.187.165.253	10.187.165.146	DNS	199	Standard query response 0xc157 A clients4.google.
613	82.996818	10.187.165.146	10.187.165.253	DNS	75	Standard query 0xf07c HTTPS ssl.gstatic.com

Frame 530: 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface \Device\NPF_{6F5A5BE-5594-4BC0-86FF-816436...}

Ethernet II, Src: 28:2e:89:cd:a8:3e (28:2e:89:cd:a8:3e), Dst: 12:18:fd:44:5a:94 (12:18:fd:44:5a:94)

Internet Protocol Version 4, Src: 10.187.165.146, Dst: 10.187.165.253

User Datagram Protocol, Src Port: 49664, Dst Port: 53

Domain Name System (query)

Source Port: 49664
Destination Port: 53
Length: 45
Checksum: 0xdf3e [unverified]
[Checksum Status: Unverified]
[Stream index: 13]
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (37 bytes)

Domain Name System (query)

0000 12 18 fd 44 5a 94 28 2e 89 cd a8 3e 00 00 45 00 ...DZ(. ...>...E
0010 00 41 66 a0 00 00 00 11 73 06 0a bb a5 92 0a bb .AF.....S.....
0020 a5 fd c2 00 00 35 00 2d df 3e c1 57 01 00 00 015...>W....
0030 00 00 00 00 00 08 63 6c 69 65 6e 74 73 34 06c lients4
0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 01 00 01 google-c om.....

3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark.

Observe how TCP retransmits lost packets while UDP does not. Explain why this

difference makes TCP reliable but slower compared to UDP. (BL4) (10)

TCP

The top screenshot shows a Wireshark capture of a TCP stream. The packet list pane displays several retransmissions of a packet with sequence number 58719. The packet details pane shows the structure of a TCP segment, including the destination port (443), sequence number (58719), and acknowledgment number (443). The packet bytes pane shows the raw data of the TCP segment.

The bottom screenshot shows a Wireshark capture of a DNS query and response. The packet list pane displays a DNS query and its corresponding response. The packet details pane shows the structure of a DNS query, including the destination port (53), source port (57887), and the query type (Standard query). The packet bytes pane shows the raw data of the DNS query.

4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

The image shows a Wireshark capture of DNS traffic. The main window displays a list of packets, with packet 90 selected. The packet details pane on the right shows the structure of the DNS packet, including the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and the Domain Name System (query) section. The query is for the domain 'settings-win.data.microsoft.com' with type A and class IN. The packet list pane shows the packet details, including the source and destination IP addresses, the protocol, and the length. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and the Domain Name System (query) section.

Wireshark - Packet 90 - Wi-Fi

Frame 90: 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface
Ethernet II, Src: 28:2e:89:cd:a8:3e (28:2e:89:cd:a8:3e), Dst: 12:18:fd:44:5a:94
Internet Protocol Version 4, Src: 10.187.165.146, Dst: 10.187.165.253
User Datagram Protocol, Src Port: 59907, Dst Port: 53
Source Port: 59907
Destination Port: 53
Length: 57
Checksum: 0x82ef [unverified]
[Checksum Status: Unverified]
[Stream index: 5]
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (40 bytes)
Domain Name System (query)
Transaction ID: 0x7642
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
settings-win.data.microsoft.com: type A, class IN
[Response In: 92]

Domain Name System: Protocol

Packets: 2934 - Displayed: 100 (3.4%) - Dropped: 0 (0.0%)

Profile: Default